

MATLAB Workbook: Numerical Solution of PDEs

From Convection to Navier--Stokes and Compressibility

Workbook Aim

This workbook introduces the numerical solution of partial differential equations using MATLAB. The emphasis is on finite-difference methods and the gradual construction of increasingly complex fluid-flow solvers.

The learning pathway is:

1D convection
↓
nonlinear convection
↓
diffusion
↓
Burgers' equation
↓
2D extension
↓
Laplace equation
↓
Poisson equation
↓
incompressible Navier--Stokes
↓
compressible flow.

Each problem introduces one or two new numerical or physical ideas.

Part A: Foundations in One Dimension

Problem 1: One-Dimensional Linear Convection

Governing equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0.$$

Here, $u(x, t)$ is the transported quantity and c is a constant wave speed.

Initial condition

Use a square-pulse initial condition:

$$u(x, 0) = \begin{cases} 2, & 0.5 \leq x \leq 1, \\ 1, & \text{otherwise.} \end{cases}$$

Boundary condition

Use

$$u(0, t) = 1.$$

Student tasks

1. Discretise the equation using forward Euler in time.
2. Use a backward difference for the spatial derivative.
3. Implement the numerical scheme in MATLAB.
4. Plot the initial condition and the final solution.
5. Vary the wave speed c and observe the effect.
6. Vary the time step Δt and observe the effect.

Questions

1. Does the pulse preserve its shape?
2. What happens when the time step is too large?
3. Does the numerical solution show smoothing?
4. What is the role of the wave speed c ?

Problem 2: CFL Condition for Linear Convection

CFL number

For the linear convection equation, the Courant--Friedrichs--Lewy number is

$$\sigma = \frac{c\Delta t}{\Delta x}.$$

Student tasks

Run the linear convection solver for the following values:

$$\sigma = 0.2, \quad \sigma = 0.5, \quad \sigma = 1.0, \quad \sigma = 1.2.$$

For each case, plot the solution after the same number of time steps.

Questions

1. Which values of σ give stable results?
2. What happens when $\sigma > 1$?
3. Why must the time step depend on the spatial grid size?
4. How does the CFL condition relate to information propagation?

Problem 3: One-Dimensional Nonlinear Convection

Governing equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0.$$

This equation differs from linear convection because the wave speed is now the solution itself.

Initial condition

Use the same square-pulse initial condition:

$$u(x, 0) = \begin{cases} 2, & 0.5 \leq x \leq 1, \\ 1, & \text{otherwise.} \end{cases}$$

Boundary condition

Use

$$u(0, t) = 1.$$

Student tasks

1. Modify the linear convection solver by replacing the constant speed c with the local value u .
2. Implement the nonlinear convection scheme.
3. Plot the solution at several time levels.
4. Compare the result with linear convection.

Questions

1. Why does the wave shape distort?
2. Which parts of the wave move faster?
3. How is nonlinear convection different from linear convection?
4. What numerical difficulties arise when the wave steepens?

Problem 4: One-Dimensional Diffusion

Governing equation

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}.$$

Here, ν is the diffusion coefficient.

Initial condition

Use the square-pulse initial condition:

$$u(x, 0) = \begin{cases} 2, & 0.5 \leq x \leq 1, \\ 1, & \text{otherwise.} \end{cases}$$

Student tasks

1. Approximate the second derivative using a central difference.
2. Implement the explicit diffusion scheme in MATLAB.
3. Plot the initial and final solutions.
4. Repeat the simulation for different values of ν .

Questions

1. What happens to sharp gradients over time?
2. How does increasing ν affect the solution?
3. Why does diffusion smooth the solution?
4. What stability restriction applies to the explicit diffusion scheme?

Problem 5: One-Dimensional Burgers' Equation

Governing equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}.$$

Burgers' equation combines nonlinear convection and diffusion.

Student tasks

1. Combine the nonlinear convection scheme with the diffusion scheme.
2. Use a square-pulse initial condition.
3. Simulate the solution for different values of ν .
4. Compare the solution with pure nonlinear convection and pure diffusion.

Questions

1. Which term causes steepening?
2. Which term causes smoothing?
3. What happens when ν is small?
4. What happens when ν is large?
5. Why is Burgers' equation a useful stepping stone toward Navier--Stokes?

Part B: Extension to Two Dimensions

Problem 6: Two-Dimensional Linear Convection

Governing equation

$$\frac{\partial u}{\partial t} + c_x \frac{\partial u}{\partial x} + c_y \frac{\partial u}{\partial y} = 0.$$

Here, c_x and c_y are constant wave speeds in the x - and y -directions.

Initial condition

Use a square pulse in a two-dimensional domain:

$$u(x, y, 0) = 2$$

inside a square region, and

$$u(x, y, 0) = 1$$

elsewhere.

Boundary conditions

Use fixed boundary values:

$$u = 1 \quad \text{on all boundaries.}$$

Student tasks

1. Create a two-dimensional grid in MATLAB.
2. Implement the finite-difference scheme for 2D linear convection.
3. Visualise the solution using surface, contour, or image plots.

- Investigate the effect of changing c_x and c_y .

Questions

- In which direction does the pulse move?
- How does two-dimensional convection differ from one-dimensional convection?
- What happens when $c_x \neq c_y$?
- How does numerical diffusion appear in two dimensions?

Problem 7: Two-Dimensional Nonlinear Convection

Governing equations

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = 0,$$
$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = 0.$$

Here, $u(x, y, t)$ and $v(x, y, t)$ are the two velocity components.

Initial condition

Use square-pulse initial conditions for both velocity components.

Boundary conditions

Use

$$u = 1, \quad v = 1$$

on all boundaries.

Student tasks

- Create two velocity fields, $u(x, y, t)$ and $v(x, y, t)$.
- Implement the coupled nonlinear convection equations.
- Plot both velocity components.
- Compare the result with 2D linear convection.

Questions

- Why are two equations needed?
- How does the u field influence the evolution of v ?
- How does the v field influence the evolution of u ?
- Why is this system more difficult than scalar convection?

Problem 8: Two-Dimensional Diffusion

Governing equation

$$\frac{\partial u}{\partial t} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right).$$

Initial condition

Use a square pulse in the centre of the domain.

Boundary conditions

Use

$$u = 1 \quad \text{on all boundaries.}$$

Student tasks

1. Implement the 2D diffusion equation.
2. Use central differences for both second derivatives.
3. Visualise the smoothing of the initial square pulse.
4. Repeat the simulation for different values of ν .

Questions

1. How does diffusion spread in two dimensions?
2. What happens near the corners of the initial square pulse?
3. How does the 2D diffusion stability condition differ from the 1D case?
4. How does increasing ν change the solution?

Problem 9: Two-Dimensional Burgers' Equations

Governing equations

$$\begin{aligned}\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= \nu \nabla^2 u, \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= \nu \nabla^2 v.\end{aligned}$$

Student tasks

1. Combine the 2D nonlinear convection scheme with the 2D diffusion scheme.
2. Simulate both velocity components.
3. Compare solutions for small and large viscosity.
4. Visualise the results using contour plots or surface plots.

Questions

1. How does the solution behave when viscosity is small?
2. How does the solution behave when viscosity is large?
3. Why does Burgers' equation resemble part of the Navier--Stokes equations?
4. What term is still missing before reaching Navier--Stokes?

Part C: Elliptic Problems and Pressure

Problem 10: Laplace Equation

Governing equation

$$\nabla^2 p = 0.$$

In two dimensions,

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 0.$$

Student tasks

1. Define a rectangular computational domain.
2. Set boundary conditions on all four sides.
3. Implement an iterative solver for the Laplace equation.
4. Continue iterations until convergence.
5. Visualise the steady solution.

Questions

1. Why is there no time derivative?
2. Why is the solution determined by boundary conditions?
3. Why is iteration required?
4. What does convergence mean for this problem?

Problem 11: Poisson Equation

Governing equation

$$\nabla^2 p = b.$$

Here, $b(x, y)$ is a source term.

Student tasks

1. Define a source term $b(x, y)$.
2. Implement the Poisson solver.
3. Compare the result with the Laplace equation.
4. Investigate how changing the source term changes the solution.

Questions

1. What role does the source term play?
2. How does Poisson's equation differ from Laplace's equation?
3. Why is the Poisson equation important for pressure calculations?
4. How does the solution depend on boundary conditions?

Problem 12: Pressure Poisson Equation

Governing equation

$$\nabla^2 p = b,$$

where b is constructed from velocity gradients.

Source term

A typical form of the source term is

$$b = \rho \left[\frac{1}{\Delta t} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial u}{\partial x} \right)^2 - 2 \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \left(\frac{\partial v}{\partial y} \right)^2 \right].$$

Student tasks

1. Given velocity fields $u(x, y)$ and $v(x, y)$, compute the source term b .
2. Solve the pressure Poisson equation.
3. Plot the resulting pressure field.
4. Investigate how the pressure changes when the velocity field changes.

Questions

1. Why does pressure depend on the velocity field?
2. How does pressure help enforce incompressibility?
3. Why is pressure not simply prescribed?
4. Why is the pressure Poisson equation solved repeatedly in Navier--Stokes simulations?

Part D: Incompressible Navier--Stokes

Problem 13: Lid-Driven Cavity Flow

Governing equations

The two-dimensional incompressible Navier--Stokes equations are

$$\begin{aligned}\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \nabla^2 u, \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \nabla^2 v.\end{aligned}$$

The incompressibility condition is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Boundary conditions

For the lid-driven cavity, the top wall moves horizontally:

$$u = U, \quad v = 0 \quad \text{on the top wall.}$$

The other walls are stationary:

$$u = 0, \quad v = 0.$$

Student tasks

1. Initialise u , v , and p .
2. Apply the velocity boundary conditions.
3. Build the pressure source term.
4. Solve the pressure Poisson equation.
5. Update the velocity fields.
6. Reapply boundary conditions.
7. Plot velocity vectors, pressure contours, and streamlines.

Questions

1. Where does the main vortex form?
2. How does viscosity affect the cavity flow?
3. How can incompressibility be checked numerically?

4. Why is the pressure Poisson equation necessary?

Problem 14: Reynolds Number Study

Reynolds number

The Reynolds number is

$$Re = \frac{UL}{\nu}.$$

Student tasks

Run the lid-driven cavity solver for several Reynolds numbers by changing ν .

Suggested values:

$$Re = 10, \quad Re = 100, \quad Re = 400.$$

Questions

1. How does the main vortex change as Reynolds number increases?
2. What happens when viscosity decreases?
3. Does the solution become harder to compute at larger Reynolds number?
4. How does the required time step change?

Problem 15: Divergence Error Check

Divergence

For incompressible flow,

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Student tasks

1. Compute the numerical divergence of the velocity field.
2. Plot the divergence field.
3. Compute a norm of the divergence error.
4. Investigate how the number of pressure iterations affects the divergence error.

Questions

1. Is the computed velocity field approximately divergence-free?
2. Where is the divergence error largest?
3. How does the pressure solver affect the divergence error?
4. Why is this a useful diagnostic?

Part E: Compressibility Extension

Problem 16: Compressible Continuity Equation

Governing equation

In one dimension, the compressible continuity equation is

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0.$$

Here, $\rho(x, t)$ is density and $u(x, t)$ is velocity.

Student tasks

1. Consider a prescribed velocity field $u(x)$.
2. Evolve the density field $\rho(x, t)$.
3. Compare the result with incompressible flow, where density is constant.

Questions

1. How does density change in compressible flow?
2. What does the quantity ρu represent?
3. Why is $\nabla \cdot \mathbf{u} = 0$ no longer the correct continuity equation?
4. What physical situations require a compressible model?

Problem 17: One-Dimensional Compressible Euler Equations

Governing equations

The one-dimensional compressible Euler equations are

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix} + \frac{\partial}{\partial x} \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix} = 0.$$

The conserved variables are

$$\mathbf{q} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}.$$

The pressure is obtained from the ideal-gas relation

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho u^2 \right).$$

Student tasks

1. Define the conservative variables.
2. Recover primitive variables from conservative variables.
3. Compute pressure using the equation of state.
4. Implement a simple conservative update.
5. Plot density, velocity, pressure, and energy.

Questions

1. Why are conservative variables used?
2. Why is an energy equation needed?
3. How is pressure obtained in compressible flow?
4. How is this different from incompressible flow?

Problem 18: Shock-Tube Problem

Initial condition

Use two constant states separated by a discontinuity.

Left state:

$$\begin{aligned}\rho_L &= 1.0, & u_L &= 0, & p_L &= 1.0. \\ \rho_R &= 0.125, & u_R &= 0, & p_R &= 0.1.\end{aligned}$$

This is a standard shock-tube initial condition.

Student tasks

1. Implement the initial discontinuity.
2. Solve the one-dimensional Euler equations.
3. Plot density, velocity, pressure, and energy.
4. Identify the main wave structures.

Questions

1. Where is the shock wave?
2. Where is the rarefaction wave?
3. Where is the contact discontinuity?
4. Why is a conservative method important for shock problems?